

Game-Theoretic Approach to LTLf Planning and Synthesis

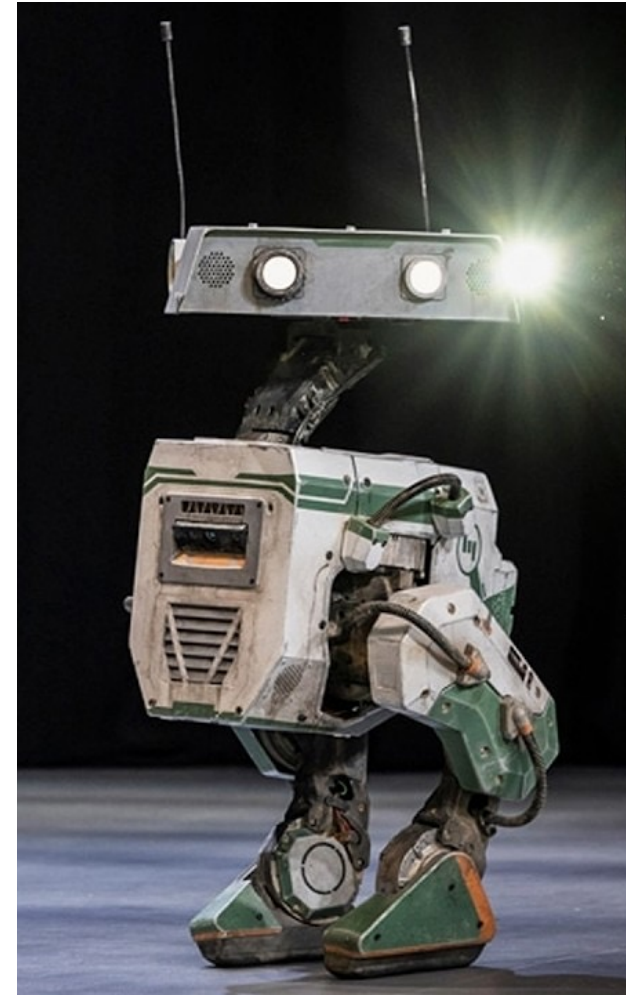
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INTRODUCTION

AI is About Building Self-Deliberating Systems!

- Autonomy is one of the grand objectives of AI.
- Aims at building autonomous agents/robots operating in changing, partially modeled, partially controllable environments.
- In other words: Aims at empowering the agent with the ability of
 - deliberating ...
 - how to act in the world ...
 - autonomously
- **First-person view:** it is the AI system (aka **agent**) that decides what to do



Autonomy and Formal Methods

WhiteMech: Whitebox Self Programming Mechanisms
ERC Advanced Grant

- Autonomy requires:
 - reasoning and planning capabilities
 - learning from experience
- Many areas of AI are concerned with autonomy:
 - Logics in AI
 - Knowledge representation and reasoning
 - Planning
 - Multi-agent systems
 - Sequential decision making (MDPs)
 - Reinforcement learning
- Recently: some objectives have been shared with automated synthesis in formal methods

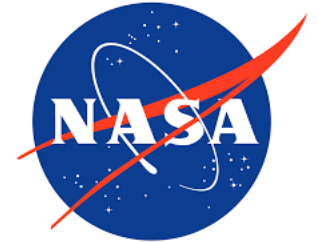


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Formal Methods

- Rigorous guarantees about the behavior of computational systems
- Wide-spread industrial adoption
- Main tasks
 - **Formalisms for specs of dynamic properties:**
 - Linear Temporal Logic
 - **Verification:**
 - Check if the system satisfies the specs.
 - **Synthesis:**
 - Synthesize a controller to make the reactive system satisfy the specs.



Reactive Synthesis

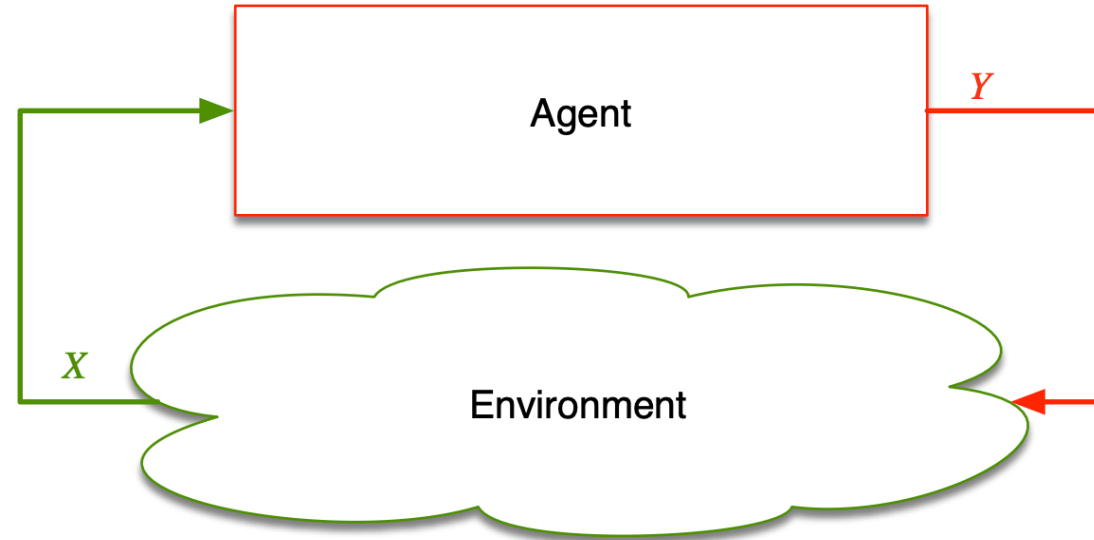
Program Synthesis

- **Basic Idea:** “Mechanical translation of human-understandable task specifications to a program that is known to meet the specifications.” [Vardi - The Siren Song of Temporal Synthesis 2018]
- Classical vs. Reactive Synthesis:
 - ▶ Classical: Synthesize transformational programs
[Green1969], [WaldingerLee1969], [Manna and Waldinger1980]
 - ▶ **Reactive:** Synthesize programs for interactive/reactive ongoing computations (protocols, controllers, robots, etc.)
[Church1963], [AbadiLamportWolper1989], [PnueliRosner1989]

Reactive Synthesis

- Reactive synthesis is equipped with a elegant and comprehensive theory
[Finkbeiner2018],[EhlersLafortuneTripakisVardi2017]
- Reactive synthesis is conceptually related to planning in nondeterministic domains
[DeGiacomoVardi2015], [DeGiacomoRubin2018], [CamachoMuisseBaierMcIlraith2018]

Reactive Synthesis in LTL



Agent strategies

Agent strategy (*also called, “plan”, “policy”, “protocol”, “process”, “program”, “behavior”*):

$$\sigma_a : (2^{\mathcal{X}})^* \rightarrow 2^{\mathcal{Y}}$$

where

- $(2^{\mathcal{X}})^*$ denotes the **history** of inputs observed so far by the agent

(a finite sequence of fluents configurations)

- $2^{\mathcal{Y}}$ denotes the **next output** of the agent

Every program/process has this form! [AbadiLamportWolper89].

Model Checking vs Synthesis

The “big bang” of the application of temporal logic to program verification: **Linear Temporal Logic** (LTL) (Pnueli, 1977)

Semantics of LTL is over ω -words, i.e., infinite traces

Note:

- ω -automata algorithms scale well as long as you do not have to determinize 
- For synthesizing controllers/strategies/policy determinization is essential 
- In AI are more interested in finite-trace semantics 

AI: Focus on Finite Traces

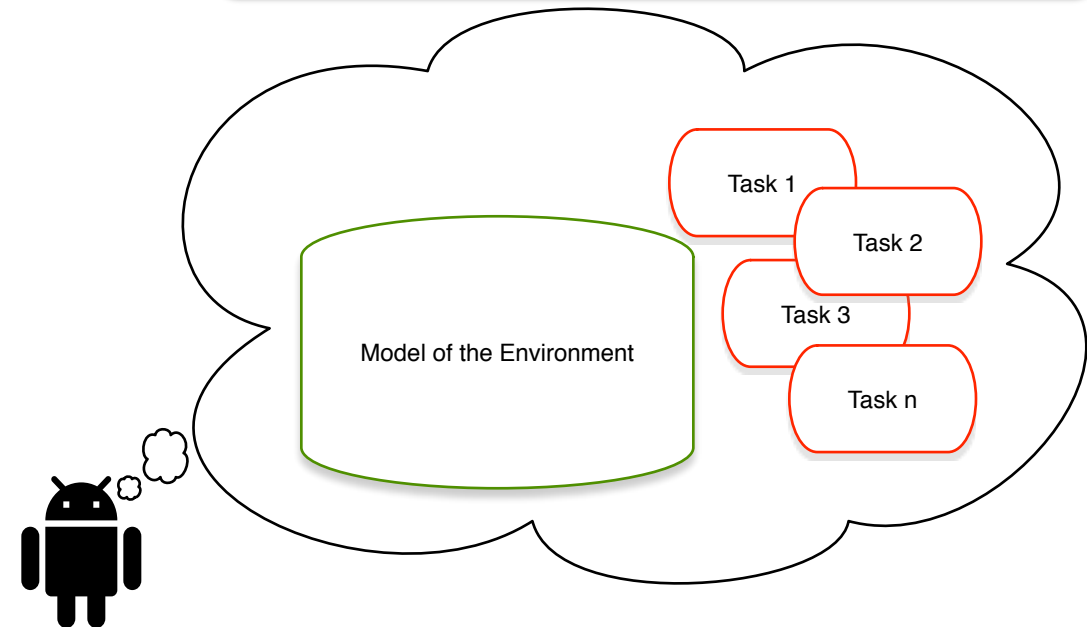
Planning in AI:

- Is all about having a task specification or “goal” and producing a “plan” (or strategy or policy or controller) to satisfy the task in the environment model.
- Which tasks?
 - A **task that terminates!**
 - Typically, just reaching a certain state in the environment

Why tasks that terminate?

- Because it is the agent that is planning/reasoning
- If the task does not terminate, the agent would be stuck doing the same task forever
- But then, why bother with equipping it with strategic reasoning abilities?
- **Note it is the agent, NOT the designer, who deliberates!**

- In *Formal Methods* focus on *infinite traces*
- In *AI* focus on *finite traces* **LTL** → **LTLf**



Synthesis in LTLf

LTLf a variant of LTL on finite traces (DeGiacomoVardi, 2013)

Semantics of LTLf is over finite (though unbounded) words, i.e., finite traces.

Reduction to regular automata

Note:

- Regular automata algorithms scale well, including determinization 
- For synthesizing controllers/strategies/policy, determinization is essential 
- In AI are more interested in finite-trace semantics 